

SUSHMITA MALLICK

1215 E Vista Del Cerro Dr, Tempe, AZ 85281

☎ 623-281-6737 ✉ sushmita@asu.edu [linkedin.com/in/sushmita01/](https://www.linkedin.com/in/sushmita01/) github.com/Sushmita01

Education

Arizona State University

Master of Science in Computer Science

GPA: 4/4, May 2024 (Expected)

Tempe, AZ

Institute of Engineering & Management

Bachelors in Electronics & Communication Engineering

GPA: 8.27/10, July 2017

WB, India

Technical Skills

Languages: Java, Python, Golang, JavaScript, C, C++, Bash, R, SQL, HTML/CSS

Machine Learning: Bayesian Theory, Data Mining, Multi-layer Neural Networks, Deep Learning, Web Crawling

Databases: MySQL, PostgreSQL, MongoDB, Redis, Aerospike, DynamoDB, S3, Neo4j

Technologies/Frameworks: Node.JS, Flask, Spring, Amazon Web Services, Apache Hive, Selenium, Kibana, WordPress

Relevant Coursework

- Statistical Learning & Pattern Recognition
- Foundations of Machine Learning
- Knowledge Representation & Reasoning
- Data Structures & Algorithms
- Data Processing at Scale
- Data Visualisation
- Operating Systems

Work Experience

ASU Decision Theater Network

Software Developer

Sept 2022 – Present

Part-time

- Using data analytics and high-performance computing to build software-integrated models which streamline big data and transform it into novel, interactive data visualizations.

Amazon

Software Development Engineer

Jun 2020 – Jun 2022

KA, India

- Worked on Amazon e-commerce website's price readjustment system, the most dynamic and fastest in the market.
- Built a web scraping engine utilizing **Java**, **Selenium** and **Javascript** which navigated dynamic web page layouts, mimicked user actions to answer on-screen questions, extracted relevant data from screen with minimal configuration.
- Developed an interactive widget for the Amazon retail website, using HTML, CSS and JavaScript for the UI, Java, Spring for the backend and AWS DynamoDB and S3 for data storage. Search "Would you like to tell us about a lower price?" on any Amazon product page to see the widget live.
- Developed a mechanism to automate the detection and rectification of unhealthy AWS server deployments - a feature that continues to save 10,000 Software Development Engineer hours per year.
- Scaled systems for peak traffic days (Black Friday) by analyzing latency metrics provisioning additional AWS servers.
- Recognized with "Tech 2022-Q1 Quarterly In it together Award" (June 2022), "Tech Annual Award for Insisting on Highest Standards" (Jan 2021) and monthly "Extra Mile" awards for Aug '20, Feb '21 and Jun '21.

Media.net

Web Application Developer

Apr 2019 – May 2020

KA, India

- Used machine learning algorithms such as Naive Bayes Classifier, Linear Regression and Gradient Boosting library XGBoost to predict user behavior (click rate) and revenue per click on website advertisements based on their geographical location, demographics, device, browser, click history etc. This helped us in targeted advertising, thus improving profit margins.
- Preprocessed data by handling missing or noisy values, reducing dimensionality and finding feature correlations.

Infosys

Systems Engineer

Dec 2017 – Apr 2019

KA, India

- Built a dashboard for a logistics and supply chain company, capable of handling real time logistics data and providing visual representation of shipments on a map.

Projects

MNIST Handwritten Digit Classification | *Statistical Machine Learning, Pattern Recognition*

Sept 2022

- Feature extraction, density estimation and bayesian classification of handwritten digits from the MNIST dataset. Built an optimal classifier to classify digits 3 and 7, achieving minimum error rate.